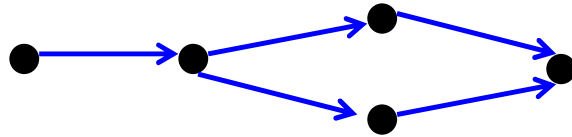
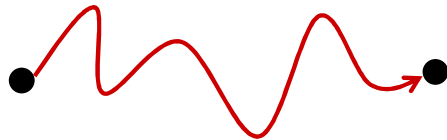


Hybrid System

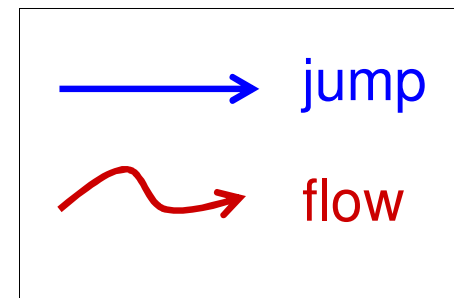
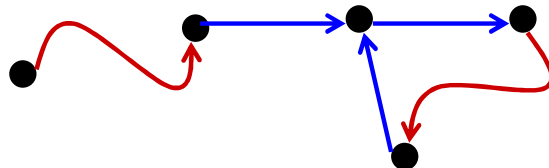
- Discrete System (FSM)



- Continuous System



- Hybrid System



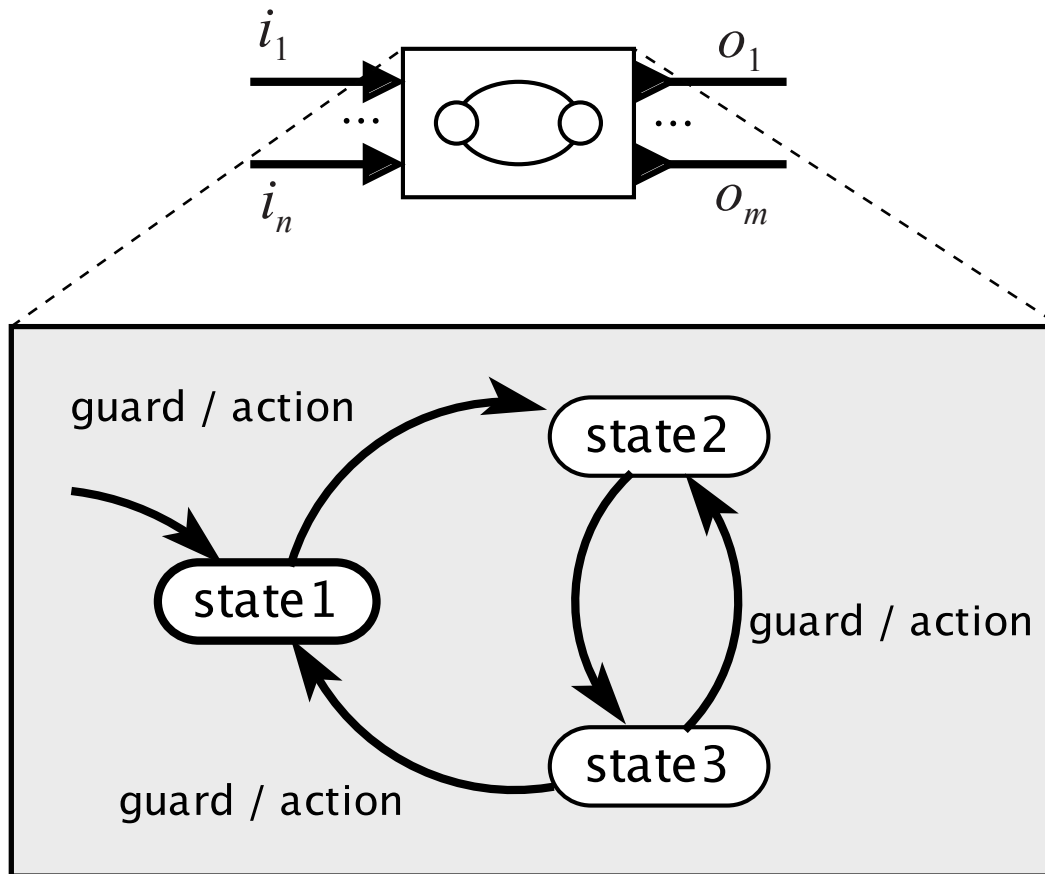
Topics

- Examples of Hybrid Systems
- The *Time Automaton* Model
 - ✦ properties of this kind of model

4.1 Modal models

- Hybrid system
 - ◆ discrete system---modeling by FSM
 - ◆ continuous system---modeling by differential and integral equations
 - ◆ hybrid system includes both discrete and continuous system
- modeling---Generalized state machine to represent combination of discrete and continuous dynamics.

4.1.1 Actor Model for State



- FSM in discrete system
- ♦ the guards on the transitions define subsets of possible values on input ports
- ♦ the actions assign values to output ports
- ♦ outputs and inputs are discrete

- hybrid system
- ♦ generalize FSMs to admit **continuous-time signals** as inputs.

4.1.2 continuous inputs

- discrete system
- ✦ FSM operates in a sequence of discrete reactions
- ✦ inputs and outputs are absent between reactions
- ✦ during the time between reactions, a state machine is understood to be **stuttering**.

4.1.2 continuous inputs

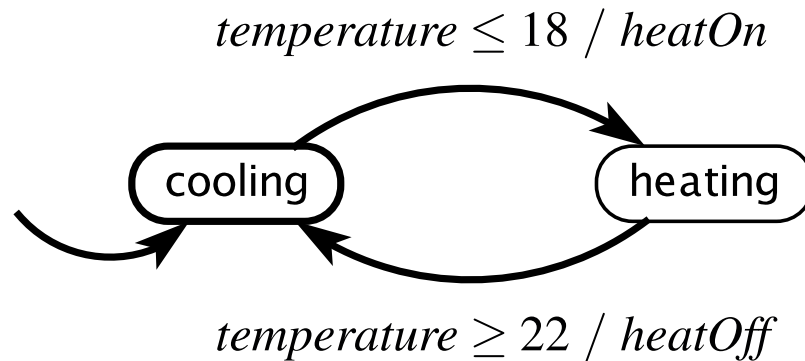
- hybrid system
- ♦ define a transition to occur when a guard becomes enabled.
- ♦ the inputs and outputs are no longer required to be absent during that time between two transitions

Example: thermostat

- for discrete system

input: $temperature : \mathbb{R}$

outputs: $heatOn, heatOff : \text{pure}$

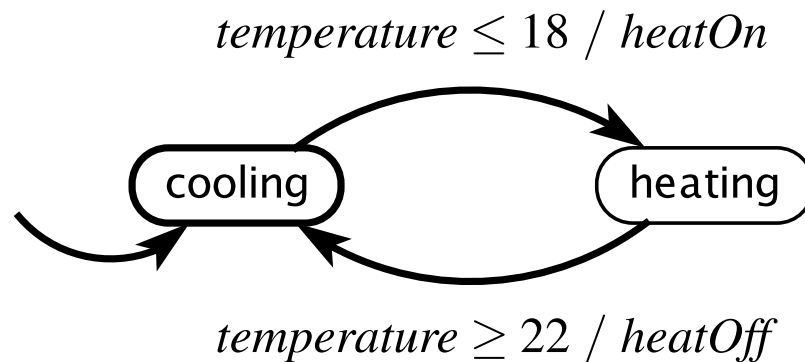


Example: thermostat

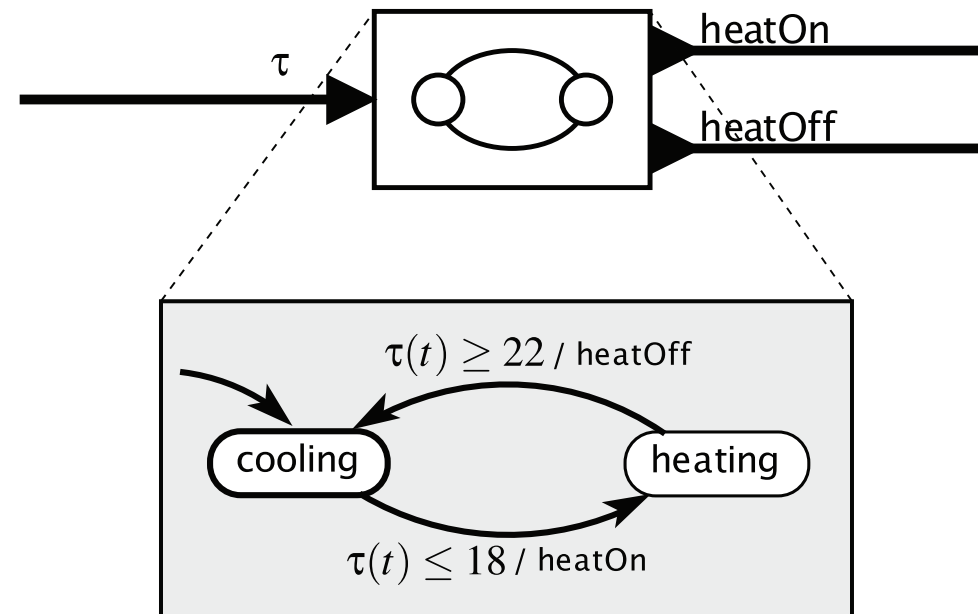
- for discrete system

input: $temperature : \mathbb{R}$

outputs: $heatOn, heatOff : \text{pure}$

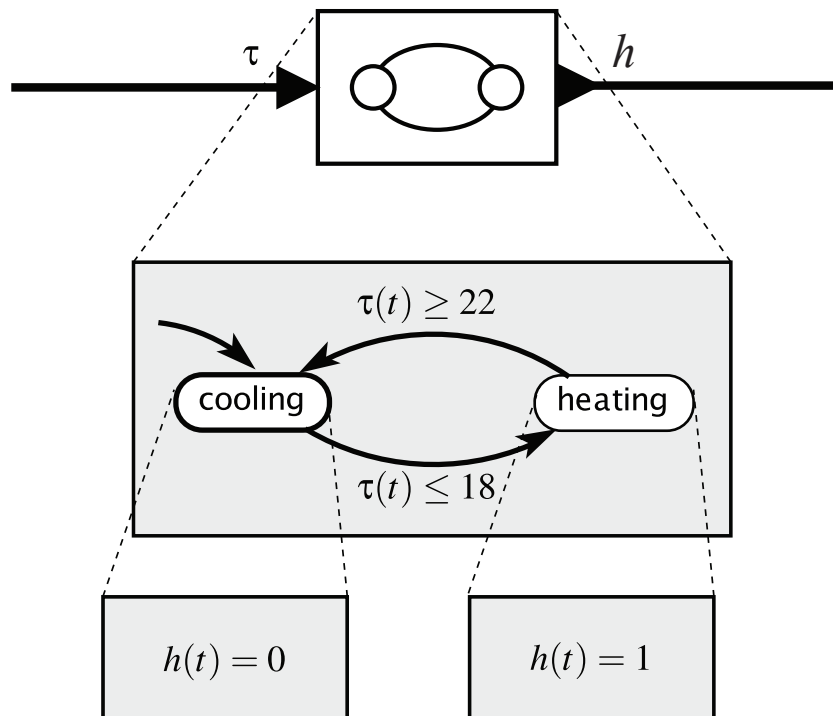


- for hybrid system
- input: $\tau: \mathbb{R} \rightarrow \mathbb{R}$



4.1.3 State Refinements

- produce a control signal whose value is 1 when the heat is on and 0 when the heat is off.



- **state refinement** gives the dynamic behavior of the output as a function of the input.

✦ Here the output is constant

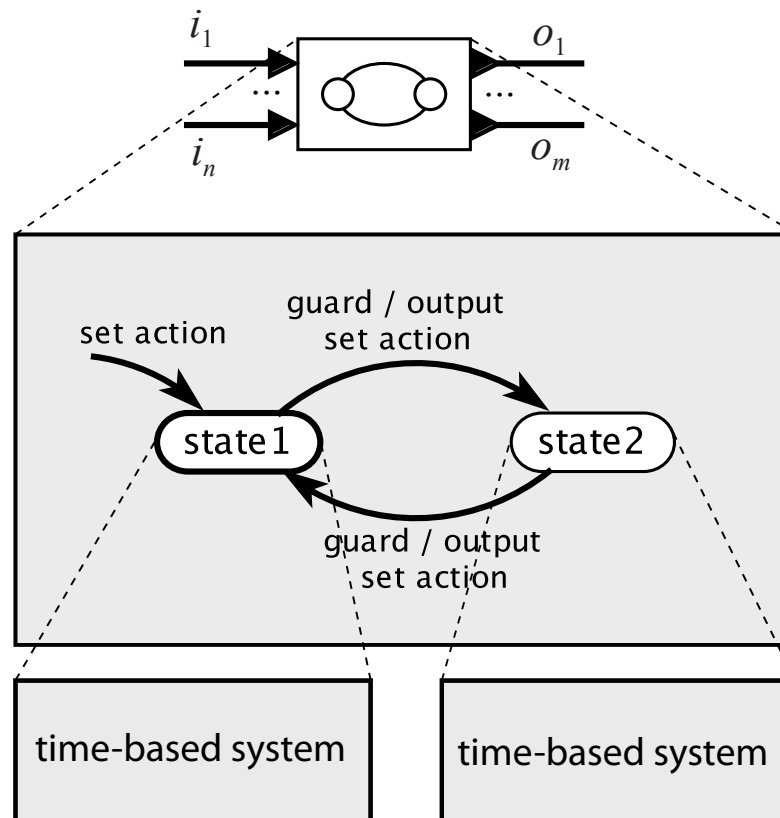
State Refinements

- Each state has a state refinement labeled as a “time-based system.”
- The state refinement defines dynamic behavior of the outputs and (possibly) additional continuous state variables.

modal model

- modal model

- ✦ a finite number of modes, one for each state of the FSM
- ✦ in a mode, it has dynamics specified by the state refinement.



4.2 Models of Hybrid Systems

4.2.1 timed automata

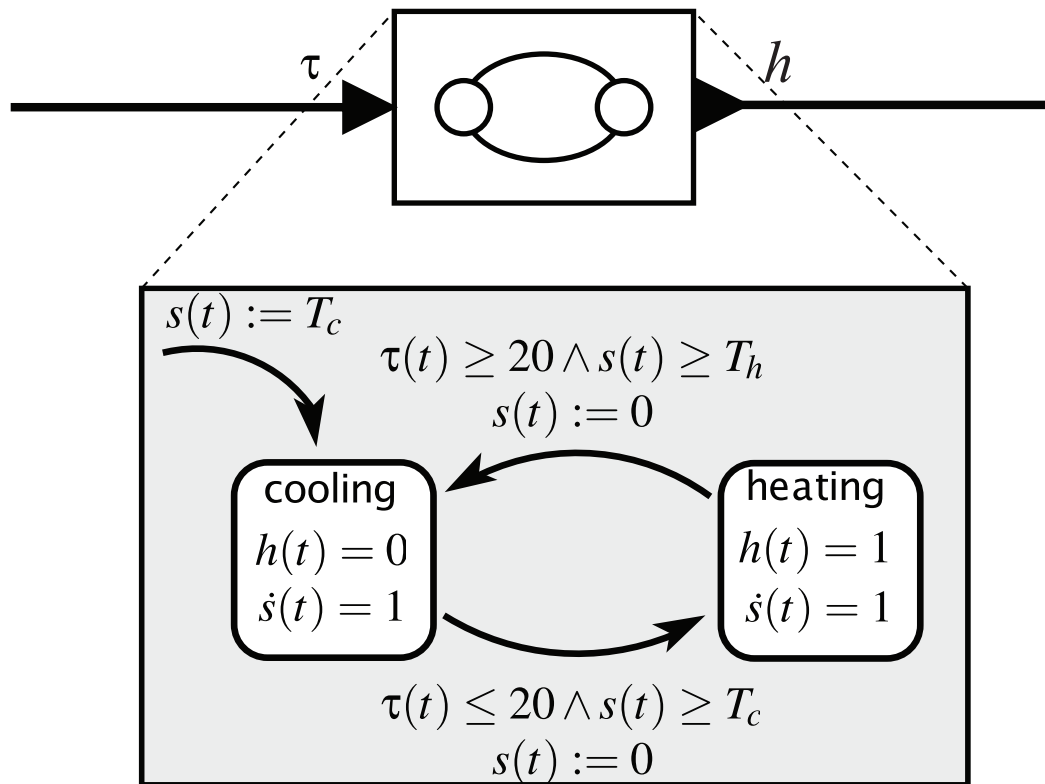
- its state progresses linearly in time.
- A clock that measures the passage of time has a simple **dynamics**
 - ♦ A clock is modeled by a first-order differential equation

$$\forall t \in T_m, \quad \dot{s}(t) = a,$$

where a is a constant

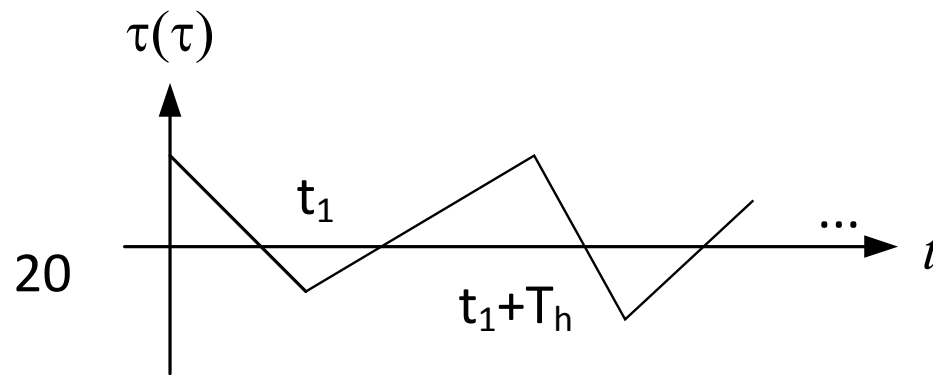
example

- Before, prevent chattering would use a single temperature threshold
- Now, the heater remain on or off for at least a minimum amount of time

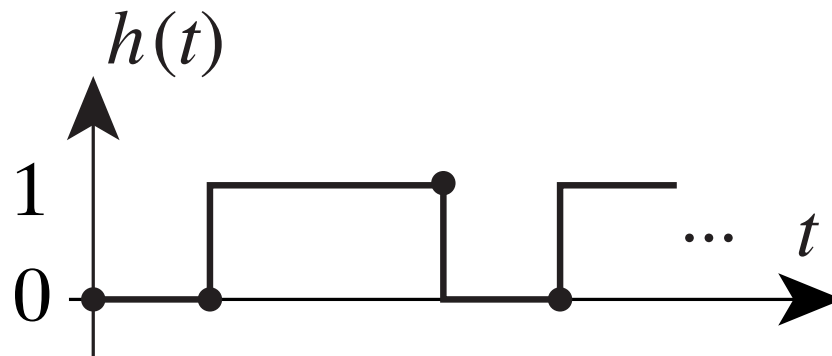


- the state refinement is shown directly within the state bubble.

Execution of this example

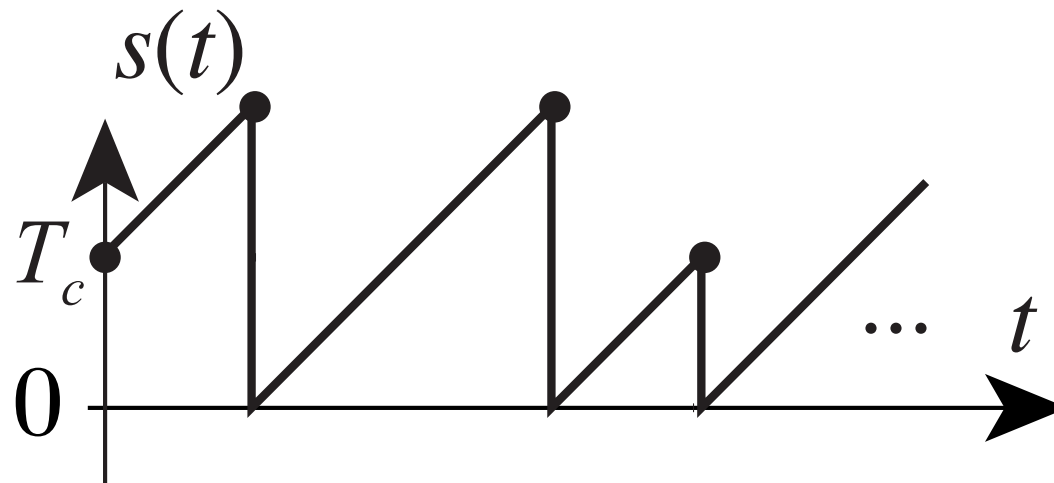


A temperature input to the hybrid system



the output h

Execution of this example



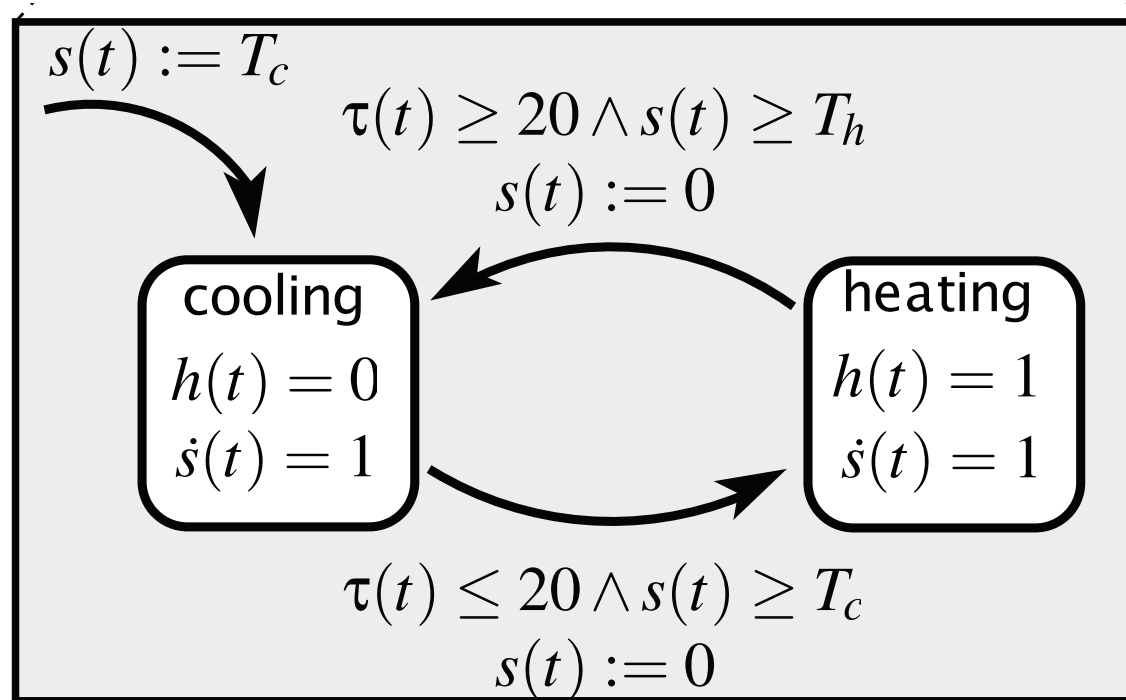
the refinement state s

Timed Automata

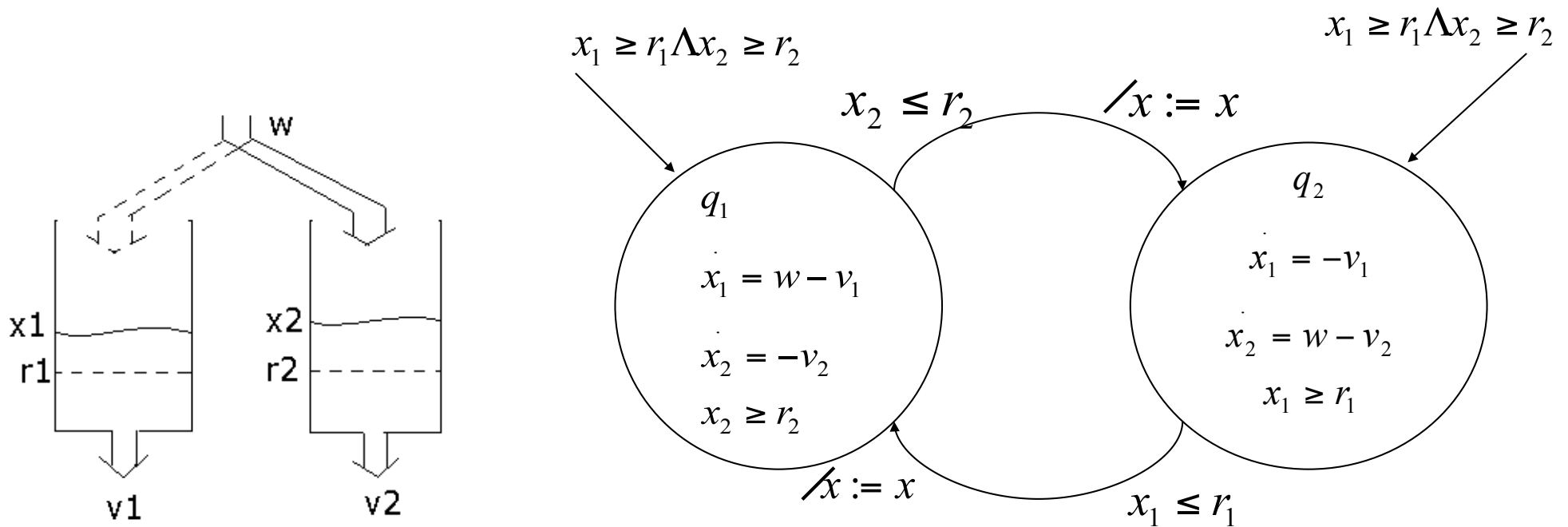
A *timed automaton* $\mathcal{A}$ is a tuple $(L, l_0, E, \text{Label}, C, \text{clocks}, \text{guard}, \text{inv})$ with

- L , a non-empty, finite set of locations with initial location $l_0 \in L$
- $E \subseteq L \times L$, a set of edges
- $\text{Label} : L \longrightarrow 2^{AP}$, a function that assigns to each location $l \in L$ a set $\text{Label}(l)$ of atomic propositions
- C , a finite set of clocks
- $\text{clocks} : E \longrightarrow 2^C$, a function that assigns to each edge $e \in E$ a set of clocks $\text{clocks}(e)$
- $\text{guard} : E \longrightarrow \Psi(C)$, a function that labels each edge $e \in E$ with a clock constraint $\text{guard}(e)$ over C , and
- $\text{inv} : L \longrightarrow \Psi(C)$, a function that assigns to each location an *invariant*.

Thermostat's timed automata



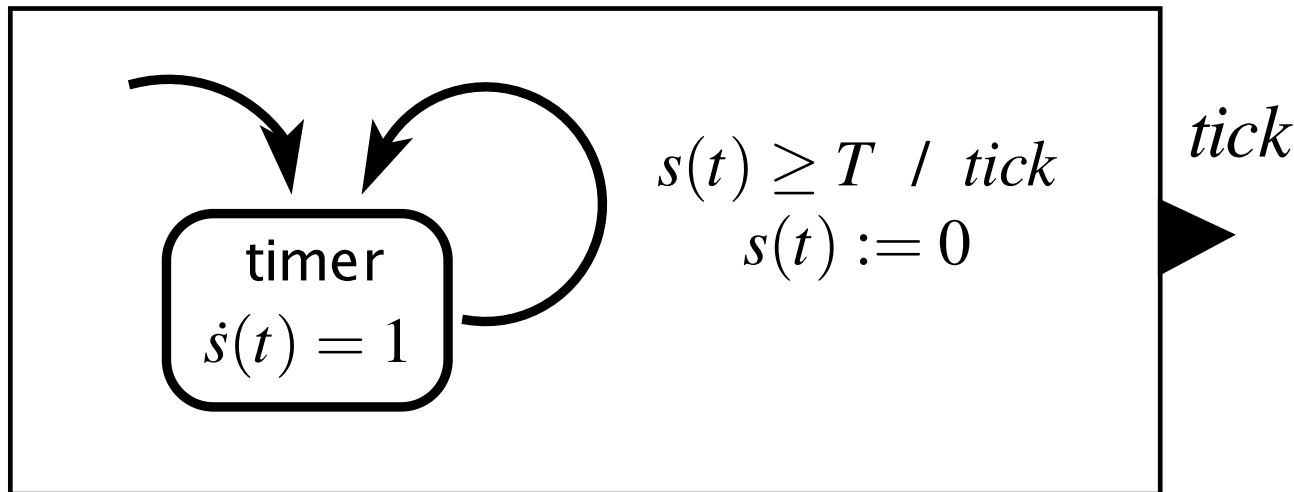
hybrid automata



Where x is a vector of x_1 and x_2

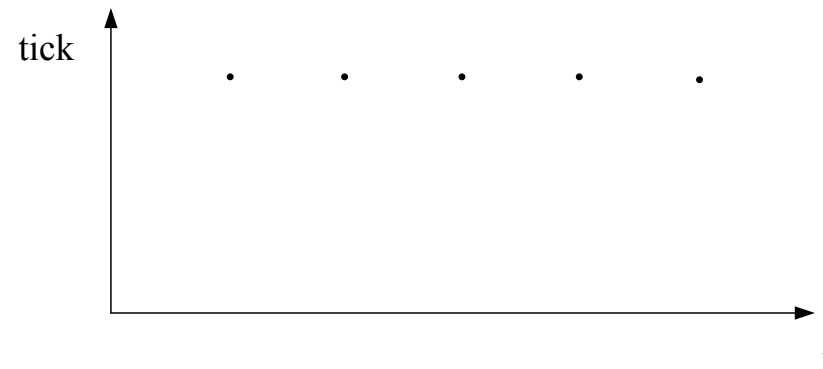
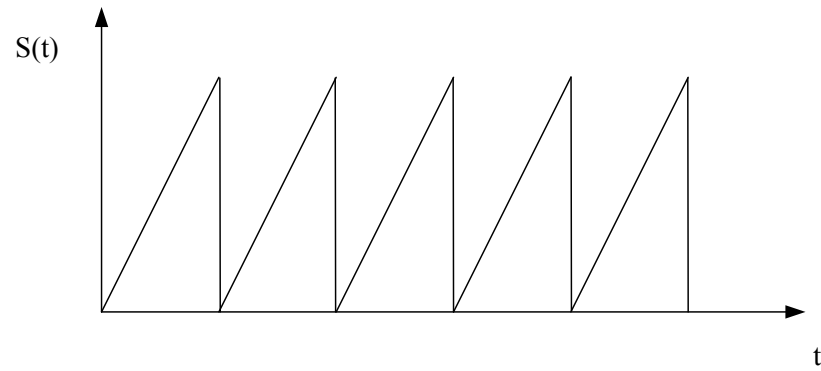
Example:timer

- transitions from one mode back to itself
- produces a pure output that will be present every T time units



- Timed automaton?

Example:timer(cont')

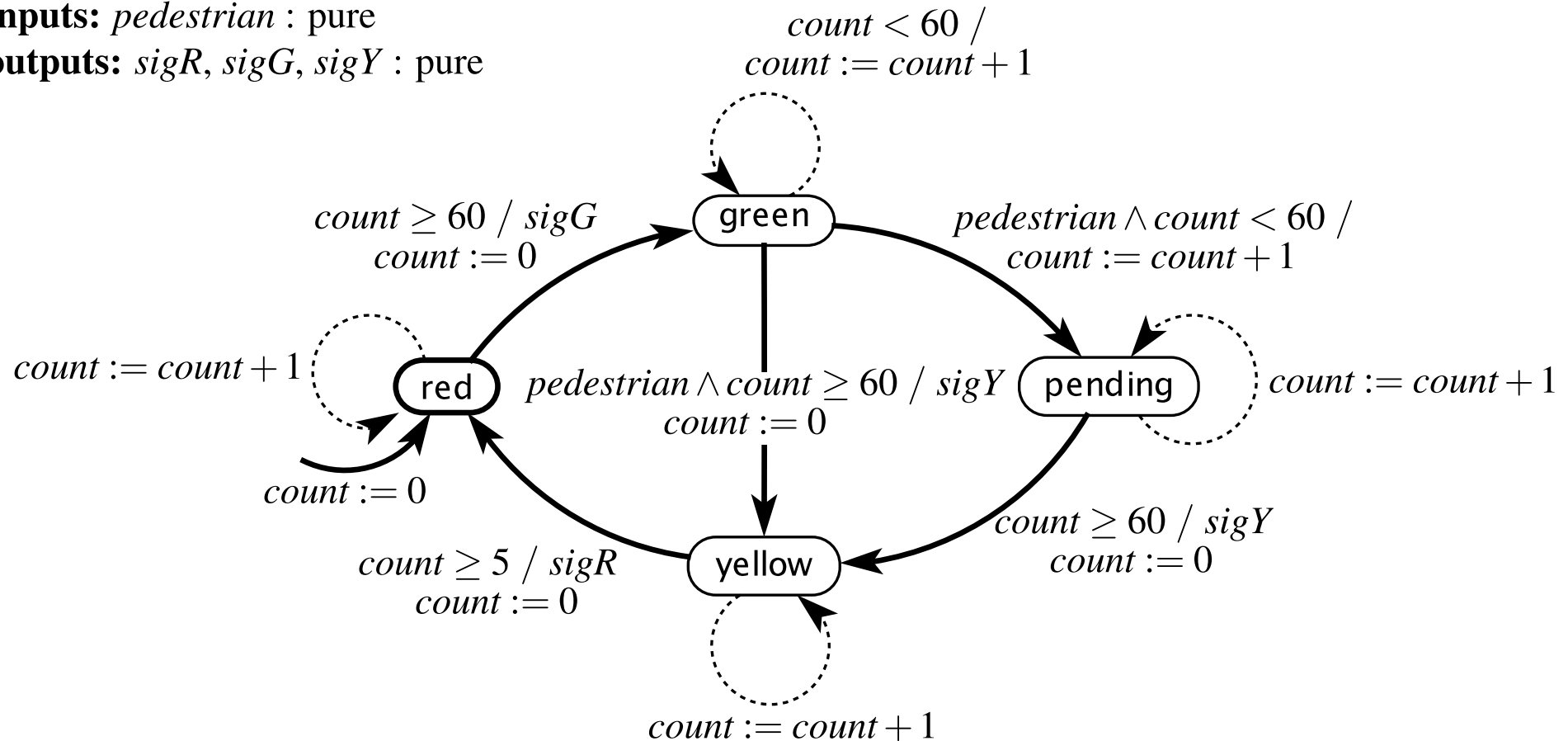


example:traffic light controller

variable: $count: \{0, \dots, 60\}$

inputs: $pedestrian : \text{pure}$

outputs: $sigR, sigG, sigY : \text{pure}$



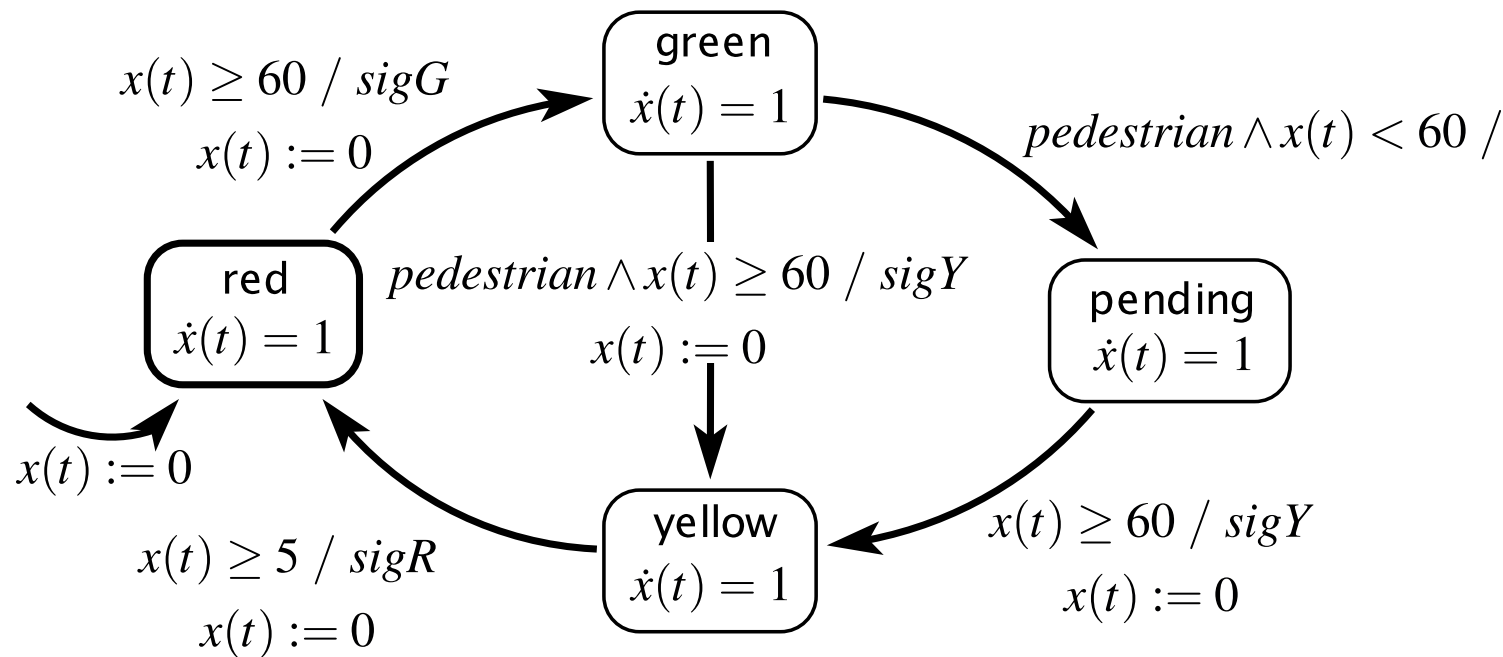
time triggered machine that assumes it reacts once each second

example:traffic light controller

continuous variable: $x(t) : \mathbb{R}$

inputs: *pedestrian*: pure

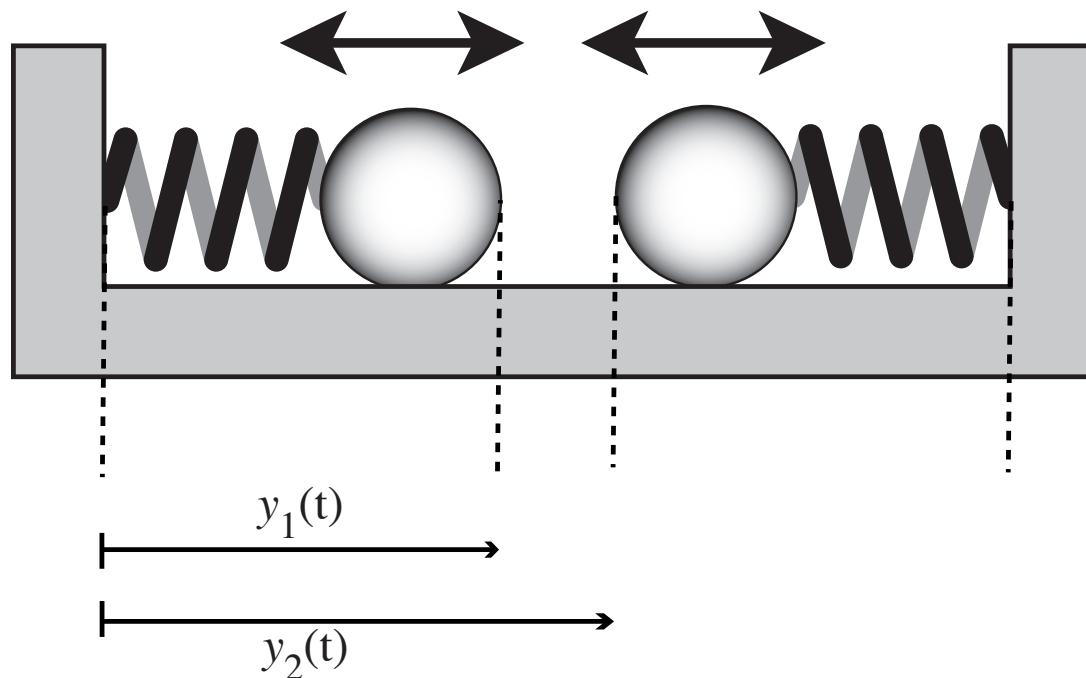
outputs: *sigR*, *sigG*, *sigY*: pure



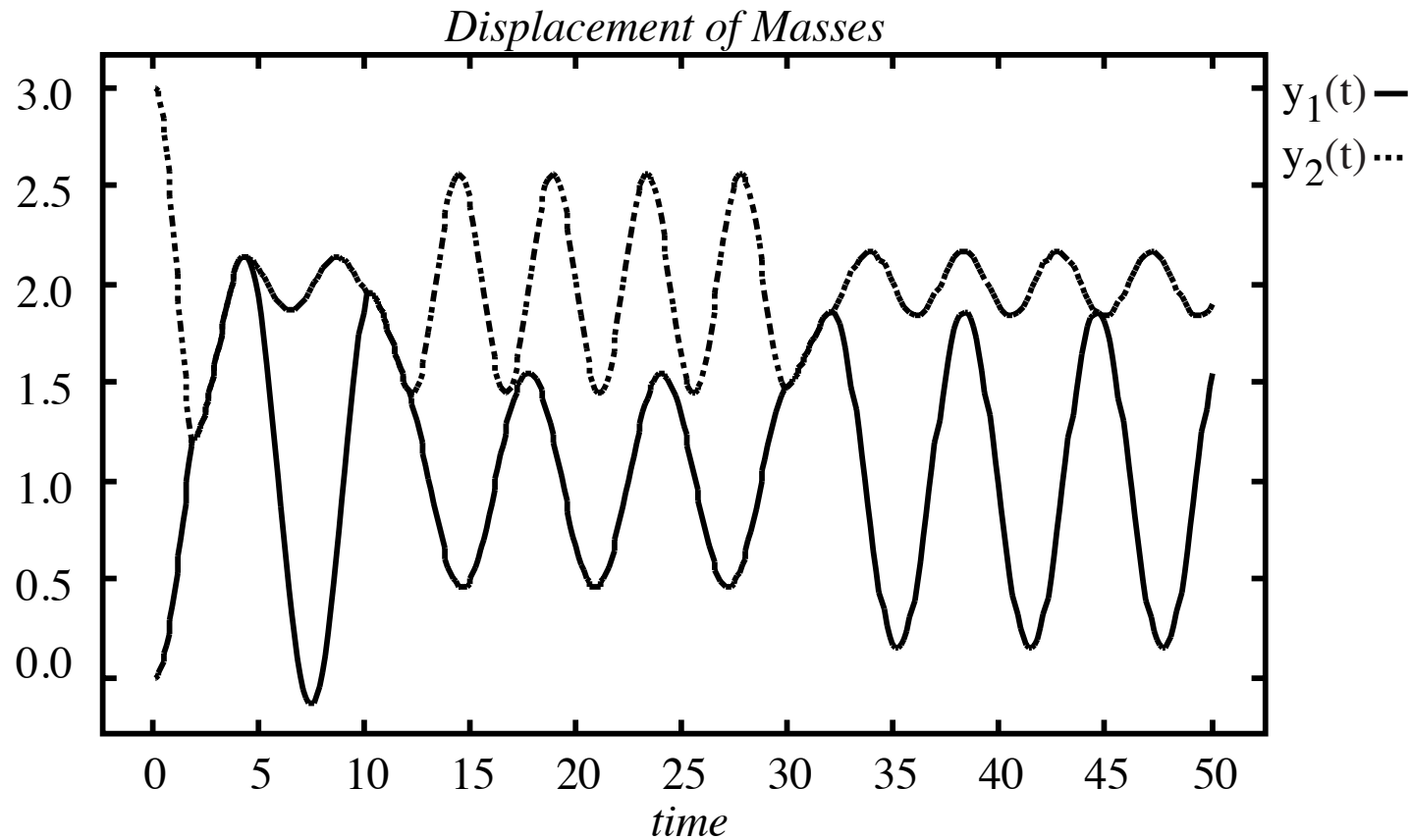
a clock represents the time passing
timed automaton??

4.2.2 Higher-Order Dynamics

- In timed automata, the time-based refinement systems is that time passes.



the trajectory of two masses



Three times collide

Sticky masses system

- For an ideal spring
 - ✦ the force at time t on the mass is proportional to $p_1 - y_1(t)$ (for the left mass) and $p_2 - y_2(t)$ (for the right mass).

$$f_1 = k_1(p_1 - y_1(t))$$

$$f_2 = k_2(p_2 - y_2(t))$$

- ✦ p_1, p_2 denote the neutral positions of the two masses

- ✦ Newton's second law

$$f = ma$$

Sticky masses system

♦ acceleration is the second derivative of the position with respect to time

$$\begin{aligned}\ddot{y}_1(t) &= k_1(p_1 - y_1(t))/m_1 \\ \ddot{y}_2(t) &= k_2(p_2 - y_2(t))/m_2.\end{aligned}$$

♦ the masses stuck together, $y_1(t)=y_2(t)$

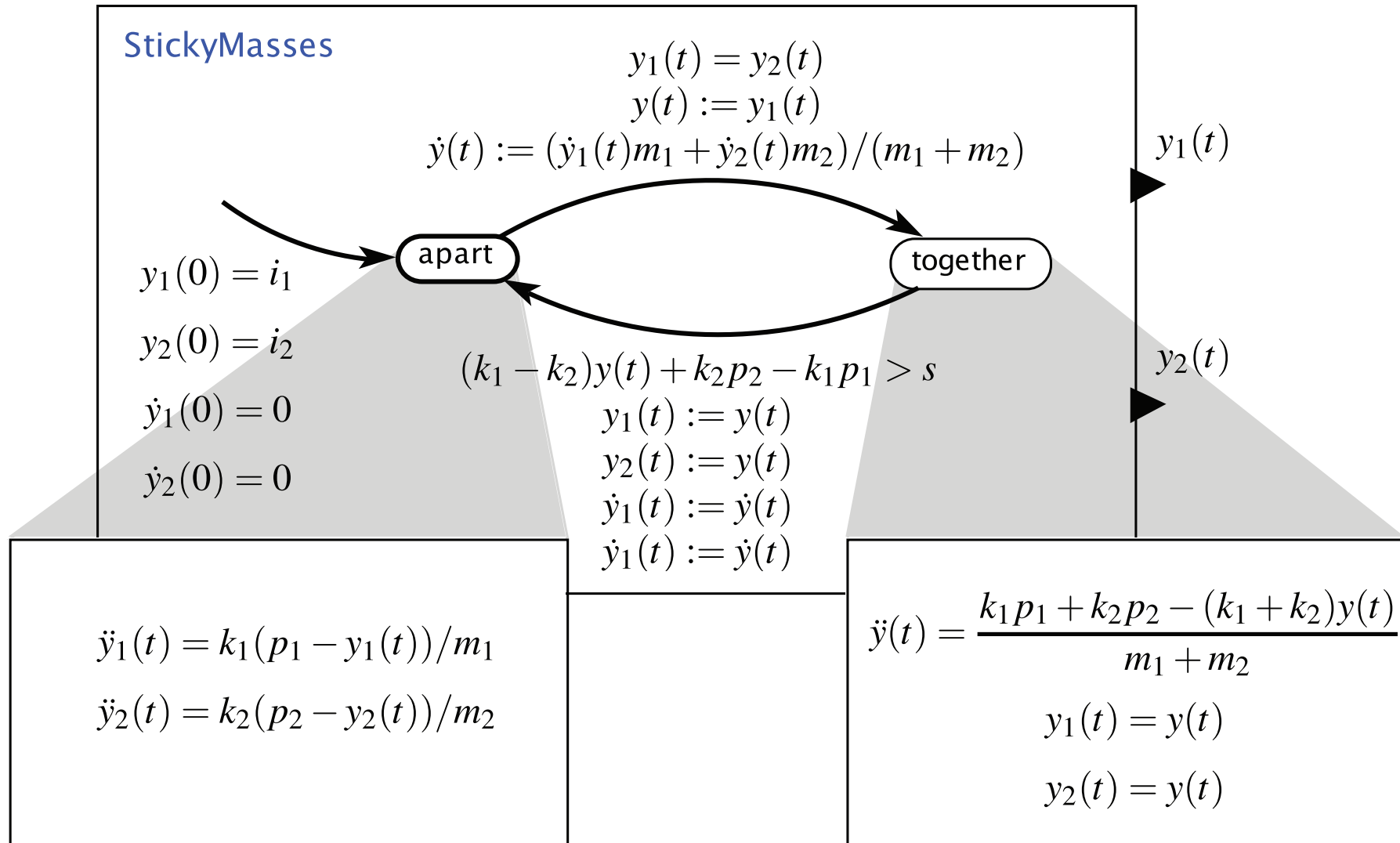
♦ they behave as a single object with mass $m_1 + m_2$.

♦ The force $f=f_1+f_2=k_1(p_1-y(t))+k_2(p_2-y(t))$

♦ the acceleration

$$\ddot{y}(t) = \frac{k_1 p_1 + k_2 p_2 - (k_1 + k_2)y(t)}{m_1 + m_2}$$

Hybrid system model



transition from **apart** to **together**

- The momentum of the left mass is $\dot{y}(t)m_1$,
- The momentum of the right mass is $\dot{y}(t)m_2$,
- The momentum of the combined masses is $\dot{y}(t)(m_1 + m_2)$.
- To make these equal,

$$\dot{y}(t) = \frac{\dot{y}_1(t)m_1 + \dot{y}_2(t)m_2}{m_1 + m_2}.$$

transition from **together** to **apart**

- s represents the stickiness of 2 masses.
- This guard is satisfied when the right-pulling force on the right mass exceeds the right-pulling force on the left mass by more than the stickiness

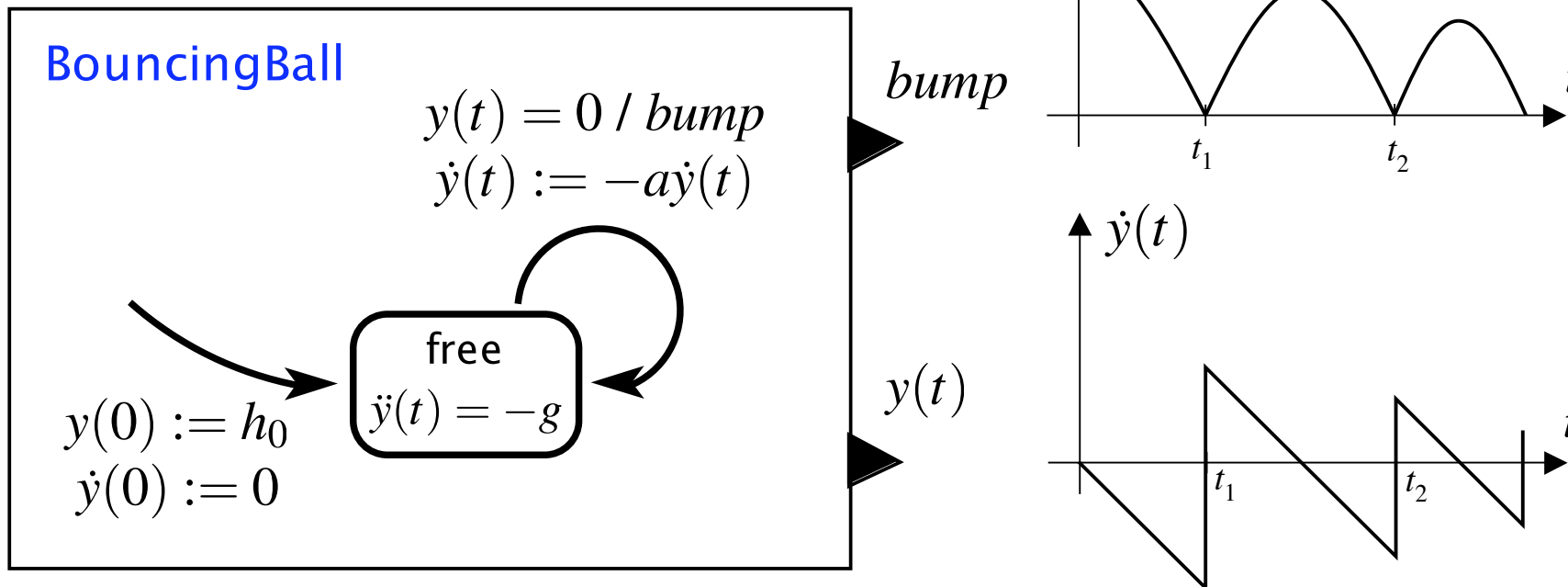
$$f_2(t) = k_2(p_2 - y(t))$$

$$f_1(t) = k_1(p_1 - y(t))$$

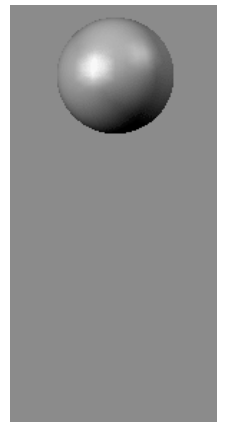
$$f_2(t) - f_1(t) = (k_1 - k_2)y(t) + k_2 p_2 - k_1 p_1 > s$$

Hybrid Automaton for Bouncing Ball

- At time $t = 0$, the ball is dropped from a height $y(0) = h_0$



Question: the time of the first hitting the ground?



Hybrid Automaton for Bouncing Ball_(cont')

$$\ddot{y}(t) = -g,$$

At the time t_1 when the ball first hits the ground, the guard

$$y(t) = 0$$

$$\dot{y}(t) = -gt,$$

$$y(t) = y(0) + \int_0^t \dot{y}(\tau) d\tau = h_0 - \frac{1}{2}gt^2.$$

$y(t_1) = 0$. It is the solution to the equation

$$h_0 - \frac{1}{2}gt^2 = 0.$$

$$t_1 = \sqrt{2h_0/g}.$$

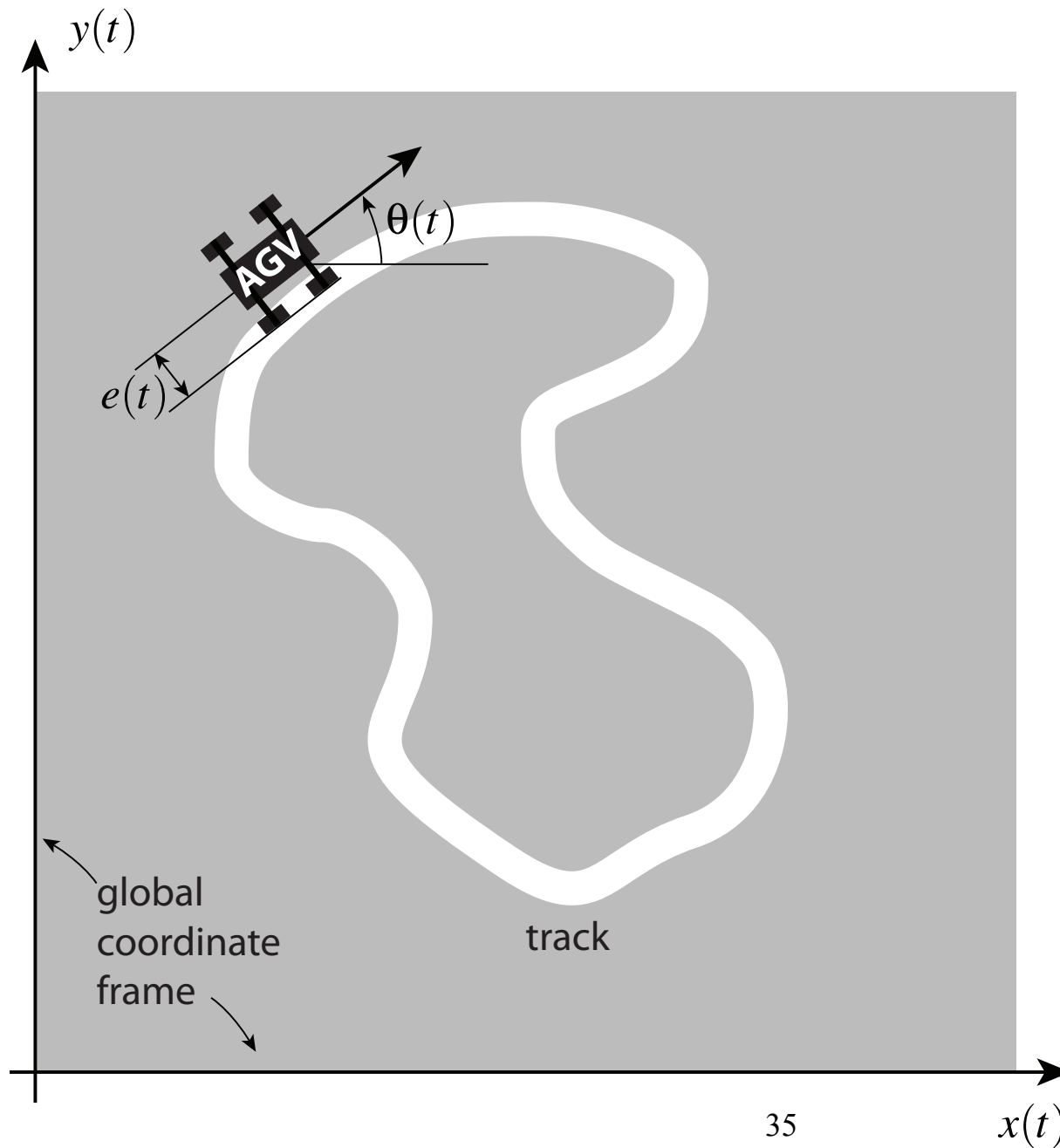
4.2.3 Supervisory control

- A control system
 - ✦ plant
 - the physical process that is to be controlled;
 - ✦ environment
 - the plant operates in the environment
 - ✦ sensors
 - measure variables of the plant and environment;
 - ✦ controller: supervisory and low-level control
 - ✓ determines the mode transition structure
 - ✓ selects the time-based inputs to the plant.

The controller

- supervisory control
 - ✦determines the mode transition structure,
- low-level control
 - ✦determines the time- based inputs to the plant.

automated guided vehicle (AGV)



- plant

$$\dot{x}(t) = u(t)\cos\theta(t),$$

$$\dot{y}(t) = u(t)\sin\theta(t),$$

$$\dot{\theta}(t) = \omega(t) .$$

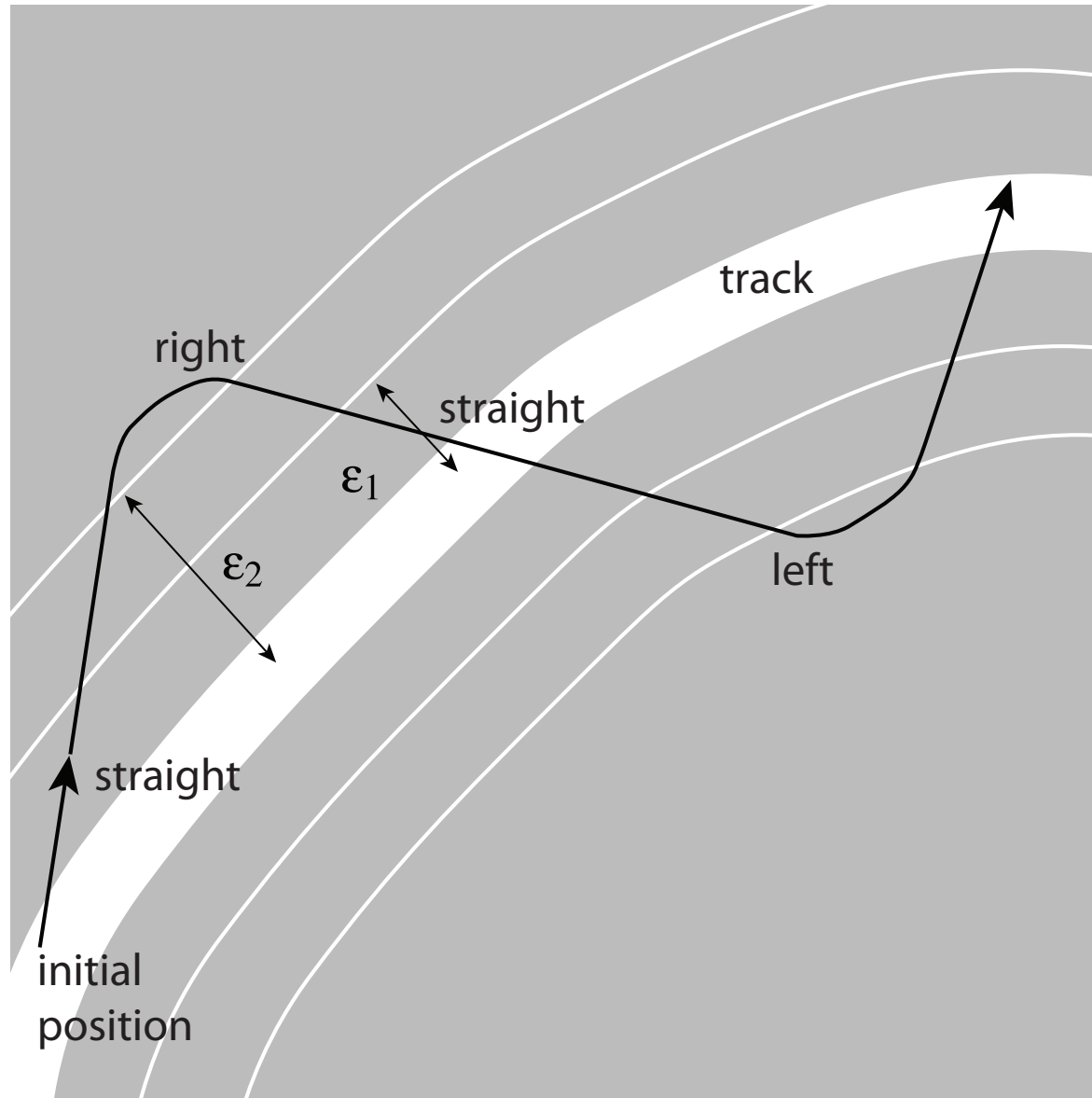
- Environment---the track

- Sensor—Measure the track of environment and the speed of the plant.

controller

- If the vehicle strays too far to the left of the track, the controller steers it towards the right
- if it strays too far to the right of the track, the controller steers it towards the left.
- If the vehicle is close to the track, the controller maintains the vehicle in a straight direction.
- the vehicle can come to a halt

A trajectory of the AGV



the low-level controller

four modes: straight, left, right, stop

straight

$$\dot{x}(t) = 10 \cos\theta(t),$$

$$\dot{y}(t) = 10 \sin\theta(t),$$

$$\dot{\theta}(t) = 0.$$

left

$$\dot{x}(t) = 10 \cos\theta(t),$$

$$\dot{y}(t) = 10 \sin\theta(t),$$

$$\dot{\theta}(t) = \pi.$$

the low-level controller

four mode: Straight, left, right, stop
right

$$\begin{aligned}x'(t) &= 10 \cos\theta(t), \\y'(t) &= 10 \sin\theta(t), \\ \theta'(t) &= -\pi.\end{aligned}$$

stop

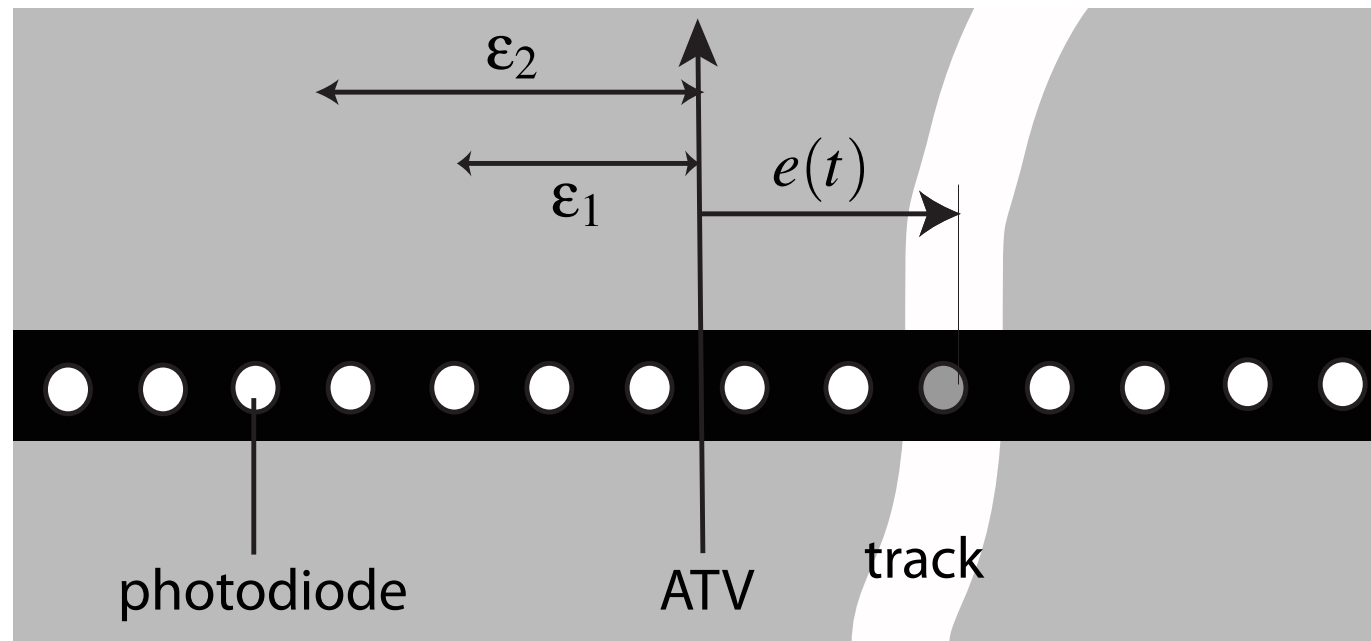
$$\begin{aligned}x'(t) &= 0, \\y'(t) &= 0, \\ \theta'(t) &= 0.\end{aligned}$$

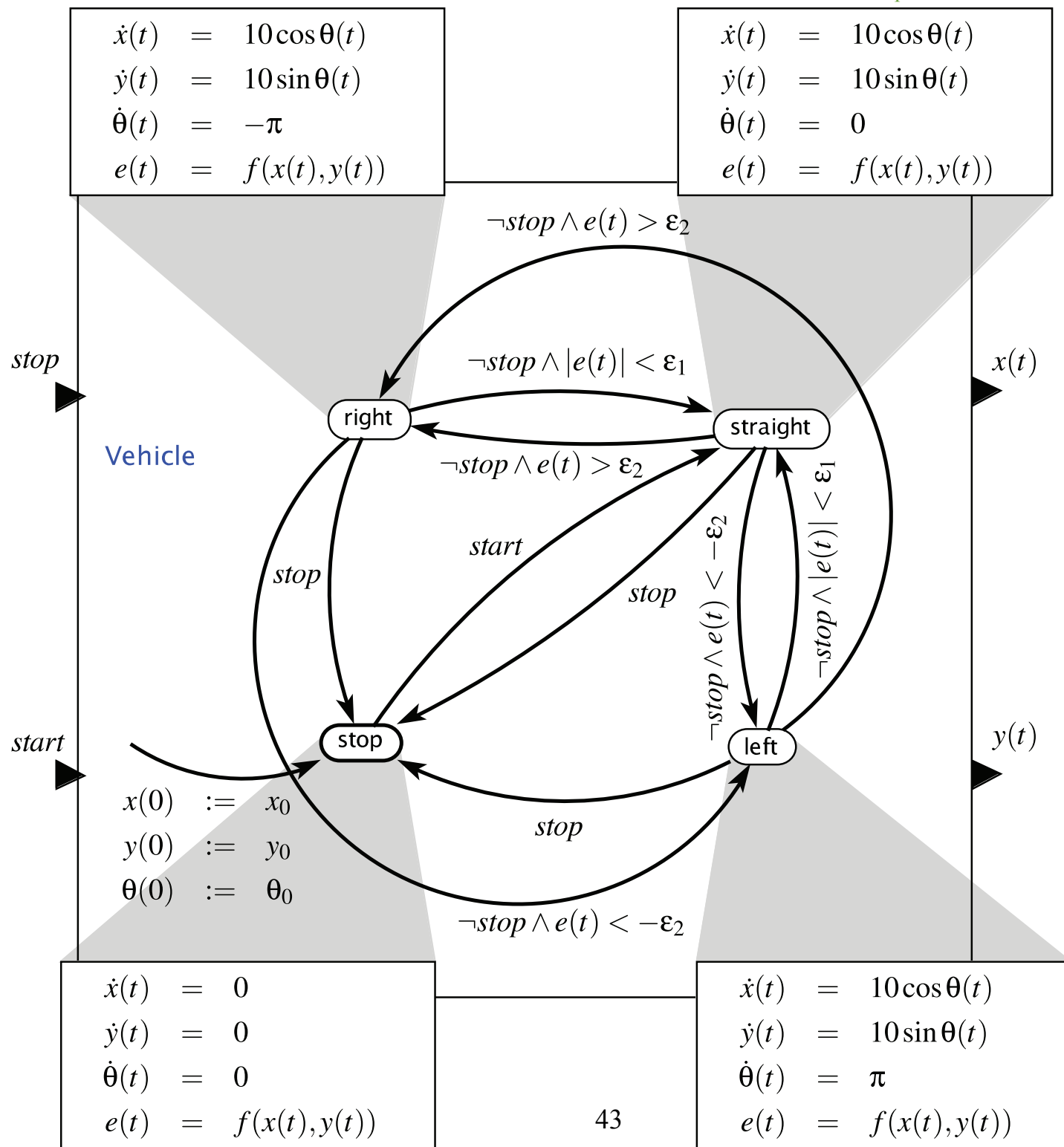
supervisory control

- governing transitions between modes
 - ✦ sensor , environment
 - ✦ how to find far away?
- the vehicle always move at its maximum speed 10mph
- when it steers the vehicle toward the opposite way, the angular velocity is the maximum speed π radians/second

sensor

- a sensor that determines how far the vehicle is to the left or right of the track





summary

- combination of time-based models and state-machine models.
- timed automata
- transition between modes describes the discrete behavior
- refinement of the mode represents the desired behavior time-based behavior.